Requirement Specification

REQ-0160 — Reliability — Battery Optimization

1. Identification

Requirement ID	REQ-0160
Title	Reliability — Battery Optimization
Type	UI Behavior
Priority	Critical
Severity	Critical
Status	Deprecated
Version	v1.1
Created	2022-08-13
Updated	2023-01-21
Source Document	SRS-APP-003-v5.pdf

2. Application & Solution

Application ID	APP-003
Application	GalaxySearch
Domain	Search
App Version	5.0.0
Solution ID	SOL-019
Solution	Media Transcoder
Component	Authentication
Feature	NFC tap-to-pay

3. Requirement Description

The system shall provide playlist shuffle securely under high-load conditions.

4. Acceptance Criteria

AC-1: The system shall satisfy the requirement within the defined performance SLA (p95 < 500ms).

AC-2: All failure scenarios must return a human-readable error message to the user.

AC-3: All events must be logged to the audit trail with a valid timestamp and user context.

AC-4: The feature must operate correctly in all supported environments (Mobile, Wearable, Cloud).

AC-5: Security constraints: all data in transit must be encrypted using TLS 1.3 or higher.

5. Dependencies

This requirement depends on the Media Transcoder (SOL-019) being operational and at version 1.4 or above.

SYNTHETIC DATA — No real Samsung confidential information

6. Related Test Cases

TC-0142	Ensure that secure key rotation does not exceed the memory c...	Smoke	Critical	Active
TC-1164	Verify that payment processing fails gracefully with a user-...	Sanity	Medium	Draft
TC-1208	Confirm that the system recovers gracefully after Data Misma...	Security	Medium	Active
TC-1632	Validate that biometric verification enforces access control...	Regression	High	Draft

7. Related Defects

BUG-0130	Network Failure in Search Indexer: SocketException: Con...	Critical	Low	OPEN
BUG-0188	Connectivity Failure in Rendering Engine: BluetoothExce...	Major	High	CLOSED

8. Synthetic Data Notice

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